

POSTACTIVATION POTENTIATION IN PROFESSIONAL RUGBY PLAYERS: OPTIMAL RECOVERY

LIAM P. KILDUFF,¹ HUW R. BEVAN,³ MIKE I.C. KINGSLEY,¹ NICK J. OWEN,¹ MARK A. BENNETT,² PAUL J. BUNCE,² ANDREW M. HORE,² JONATHAN R. MAW,¹ AND DAN J. CUNNINGHAM¹

¹Department of Sports Science, Vivian Tower, University of Wales Swansea, Swansea SA2 8PP, United Kingdom;

²Welsh Rugby Union, The Barn, Vale of Glamorgan CF72 8JY, United Kingdom; ³Ospreys Rugby, New Stadium, Landore, Swansea SA1 2FA, United Kingdom.

ABSTRACT. Kilduff, L.P., H.R. Bevan, M.I.C. Kingsley, N.J. Owen, M.A. Bennett, P.J. Bunce, A.M. Hore, J.R. Maw, and D.J. Cunningham. Postactivation potentiation in professional rugby players: Optimal recovery. *J. Strength Cond. Res.* 21(4):1134–1138. 2007.—Following a bout of high-intensity exercise of short duration (preload stimulus), the muscle is in both a fatigued and a potentiated (referred to as postactivation potentiation) state. Consequently, subsequent muscle performance depends on the balance between these 2 factors. To date, there is no uniform agreement about the optimal recovery required between the preload stimulus and subsequent muscle performance to gain optimal performance benefits. The aim of the present study was to determine the optimal recovery time required to observe enhanced muscle performance following the preload stimulus. Twenty-three professional rugby players (13 senior international players) performed 7 countermovement jumps (CMJs) and 7 ballistic bench throws at the following time points after a preload stimulus (3 repetition maximum [3RM]): baseline, ~15 seconds, and 4, 8, 12, 16, and 20 minutes. Their peak power output (PPO) was determined at each time point. Statistical analyses revealed a significant decrease in PPO for both the upper (856 ± 121 W vs. 816 ± 121 W, $p < 0.001$) and the lower ($4,568 \pm 509$ W vs. $4,430 \pm 495$ W, $p = 0.005$) body when the explosive activity was performed ~15 seconds after the preload stimulus. However, when 12 minutes was allowed between the preload stimulus and the CMJ and ballistic bench throws, PPO was increased by $8.0 \pm 8.0\%$ and $5.3 \pm 4.5\%$, respectively. Based on the above results, we conclude that muscle performance (e.g., power) can be significantly enhanced following a bout of heavy exercise (preload stimulus) in both the upper and the lower body, provided that adequate recovery (8–12 minutes) is given between the preload stimulus and the explosive activity.

KEY WORDS. contrast training, power development, motor unit excitability, myosin light chain

INTRODUCTION

The contractile history of skeletal muscle plays a pivotal role in the subsequent performance from that muscle group. For example, following a bout of high-intensity exercise of short duration (e.g., a preload stimulus), subsequent muscle performance (e.g., explosive activity) has been demonstrated by some researchers to be decreased (9, 11, 18, 19, 21–23), which can be attributed to muscle fatigue (e.g., low intramuscular stores of phosphocreatine) associated with the preload stimulus. In contrast to this finding, researchers have also reported increases in performance during subsequent muscle performance following a preload stimulus (3, 4, 6, 7, 10, 24–26). This increase in muscle performance has been attributed to a condition referred to as postactivation potentiation (PAP) (e.g., 13). To date, there has been much speculation regarding the potential mechanisms responsible for PAP and the subsequent improve-

ment in performance; nevertheless, increased phosphorylation of the regulatory myosin light chain at the molecular level and increased reflex electrical activity in the spinal cord are the most plausible mechanisms (1).

Based on the above, it is clear that both fatigue and PAP can coexist in skeletal muscle (20) and that muscle performance following a preload stimulus depends on the balance between muscle fatigue and muscle potentiation (5).

In recent years, there has been an abundance of studies examining the effectiveness PAP on subsequent power output in both upper and lower body with conflicting results (3, 4, 6, 7, 9, 10, 18, 19, 21–26). For example, Baker (3) examined the effectiveness of PAP on upper-body power development and found that the power output was 4.5% greater in a bench throw activity performed 3 minutes after the preload (5 repetition maximum [5RM]) compared to a bench throw performed without any preload. This finding has been verified by numerous researchers (e.g., 6, 26). However, as previously stated, some studies have reported no effect or even a slight decrease in power outputs followed the preload stimulus (e.g., 5, 9, 11, 18, 19).

This conflict in the literature may in part be explained by numerous methodological differences in the various studies (e.g., training status of subjects) (16). The majority of the methodological limitations mentioned in the Hodgson et al. (16) review paper can be overcome by careful study design; however, to date, there is no uniform agreement about the optimal recovery time between the preload stimulus and subsequent explosive activity with studies reporting recovery periods ranging from 0 to 18.5 minutes (3, 5, 6, 8, 12, 26). To date, only 1 study has attempted to examine the optimal recovery time between the preload stimulus and subsequent explosive activity, with Jensen and Ebben (18) examining recovery periods of 10 seconds and 1, 2, 3, and 4 minutes between the preload stimulus (squats) and the subsequent explosive activity (countermovement vertical jump). They concluded that there was no effect on jump performance following the preload stimulus at any of the specified recovery periods; however, they suggested that greater than 4 minutes of recovery may be needed to see a performance enhancement (18).

Therefore, in light of the above, the aim of the present study was to determine the optimal recovery time for maximal benefits between the preload stimulus and the explosive activity in a group of professional rugby players.

METHODS

Experimental Approach to the Problem

During this within-subject design study, each subject was required to attend the laboratory on at least 3 occasions.

TABLE 1. Physical characteristics of subjects at baseline ($n = 23$).*

Variables	Mean $\pm$ SD
Weight (kg)	97.4 $\pm$ 13.4
Stature (cm)	185 $\pm$ 8
Age (y)	24.0 $\pm$ 3.4
1RM squat (kg)†	153 $\pm$ 23
1RM bench (kg)†	124 $\pm$ 16
Professional rugby experience (y)	4.0 $\pm$ 3.4

* 1RM = 1 repetition maximum

† Estimated from 3RM.

The objective of the first testing session was to determine the subjects' 3RM on the squat and bench press; in addition, the subjects were familiarized to the study procedures that were to follow. During the 2 experimental days, subjects completed a baseline explosive activity (countermovement jump [CMJ] or ballistic bench press), then, following a 10-minute recovery period, subjects were required to complete a preload stimulus (3RM) on the squat or bench press. Immediately following the preload stimulus (within 15 seconds) and every 4 minutes after the preload stimulus up to 20 minutes (4, 8, 12, 16, and 20 minutes), the subjects completed either a CMJ or a ballistic bench throw.

Subjects

Twenty-three professional rugby players (Table 1) from whom written informed consent had been obtained volunteered to take part in the present study, which was approved by the local ethics committee and carried out during the preseason (August–September). No subject had a record of cardiovascular or respiratory disease, and none had subjective evidence of musculoskeletal injury. Subjects were recruited on the basis that they were engaged in a structured weight-training program for at least 1 year prior to the start of the study and were able to complete the bench press and squat movements with correct technique. At the time of entry into the study, subjects had completed a power phase, which incorporated Olympic lifts, the various derivatives of the Olympic lifts, and complex/contrast exercises. The average resistance training experience of the present group of subjects was 3.1 ± 1.6 years. The subjects' eligibility was assessed by interview prior to their informed consent for participation in the study.

Experimental Procedures

Prior to the commencement of the main experimental trials, subjects visited the laboratory in order to become familiar with the testing methods and to have their 3RM bench press and squat measured. During the familiarization trial, subjects practiced performing CMJs and ballistic bench presses; in addition, all subjects in the present study had participated in an optimal loading experiment in the past and were therefore well familiarized with both testing methods. Following the familiarization and strength testing period, all subjects performed 2 tests (upper- and lower-body contrast exercise). The first test was conducted 48 hours after the familiarization and strength trial, while the second test was conducted 48 hours after the first testing session. Subjects reported to the laboratory on the morning of testing after having refrained from alcohol, caffeine, and strenuous exercise the day before. Following the measurement of each subject's stature and body mass, subjects underwent a standardized warm-up

consisting of 5 minutes of light-intensity cycling, followed by a series of dynamic stretches with an emphasis on stretching the musculature associated with the squat and CMJ or the bench press and bench throw, depending on the testing session.

On test day 1 following the standardized warm-up, subjects completed a body mass baseline CMJ. Following a 10-minute recovery period, subjects completed their preload stimulus on the squat. Immediately following the preload stimulus (within 15 seconds) and every 4 minutes after the preload stimulus up to 20 minutes (4, 8, 12, 16, and 20 minutes), the subjects completed a CMJ.

On test day 2, upper-body testing was carried out at the same time of day and in the same manner as lower-body testing with the exception that ballistic bench throws and 3RM bench press replaced CMJ and squats as the modes of exercise. Consumption of water (500 ml) was permitted during each test. Room temperature was maintained between 20 and 24°C. Verbal encouragement was given to maximize performance.

Measurements

Strength Testing. Prior to the start of the strength testing session, all subjects underwent a standardized warm-up that consisted of light-intensity cycling for 5 minutes, followed by a series of dynamic movements with an emphasis on warming up the musculature associated with the squat or bench press. Subjects then performed 3 warm-up sets of 10 repetitions at 50% of their estimated 1RM. Following the warm-up set, subjects attempted 3 repetitions of a set load, and if successful, the lifting weight was increased until the subject could not lift the weight through the full range of motion. All subjects had been previously exposed to 3RM testing for both the bench press and the squat. A 5-minute rest was imposed between all attempts to allow subjects adequate time to replenish energy stores. The 3RM was determined after 2–3 attempts in all subjects. Upper-body strength testing was carried out in the same manner. Both the bench press and the squat movements were carried out as per the International Powerlifting Federation rules (17).

Upper- and Lower-Body Power Output. Upper-body peak power output (PPO) was tested during a ballistic bench press throw performed on a Smith machine with a relative resistance of 40% of their predicted 1RM (40% 1RM has previously been shown to be the optimal load for developing PPO in the upper body in rugby players in our laboratory). During each bench press throw, the subject was instructed and encouraged to lift the bar from the starting position and throw it as high as possible. To avoid the effects of deceleration and achieve maximal bar velocity, the bar was released at the top of the range of motion. During each throw, subjects were required to keep their head, shoulders, and trunk in contact with the bench as well as their feet in contact with the floor.

Subjects completed 7 ballistic bench throws at the following time points: baseline, immediately after the preload stimulus (~15 seconds), and every 4 minutes until 20 minutes (4, 8, 12, 16, and 20 minutes). The preload stimulus consisted of 1 set of 3RM on the bench press, which was performed 10 minutes after the baseline ballistic bench throw.

For the measurement of lower-body PPO, subjects completed CMJs with a broomstick over their shoulder in a squat position (used to attach Ballistic Measurement System [BMS]; Fitness Technology, Skye, South Australia, Australia) with the subjects squatting down to a pre-

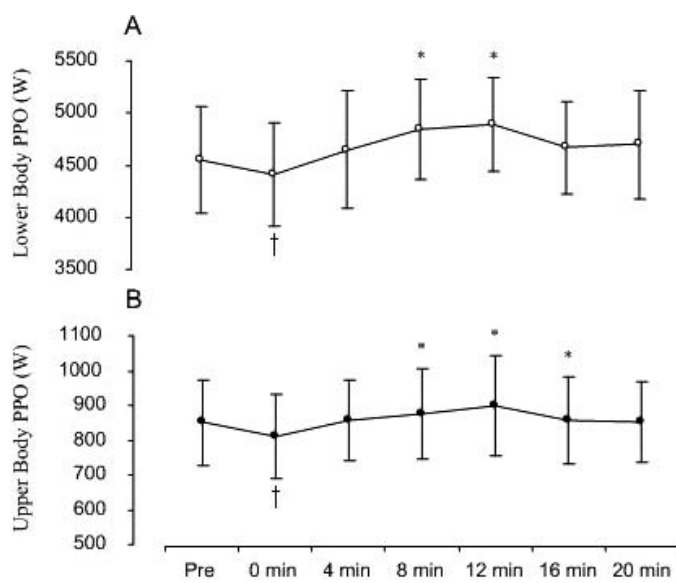


FIGURE 1. Peak power outputs (PPO) across 7 time points relative to body mass for the lower (A) and upper (B) body.

determined depth and explosively jumping to the highest height attainable. Again subjects completed 7 jumps at the same time as those recorded for the ballistic bench throws (baseline, ~15 seconds, and 4, 8, 12, 16, and 20 minutes). The preload stimulus consisted of 1 set of 3RM on the squat, which was performed 10 minutes after the baseline CMJ.

Ballistic Measurement System. The PPO from both the CMJs (including body mass in the calculation) and the ballistic bench throws was calculated using the software provided with the BMS. The BMS was used to collect bar displacement data during both the jump squats and the ballistic bench throws. The BMS comprises a cable-extension potentiometer (distance transducer) that produces a variable-voltage output in relation to the extension of the 3-m cable. An analog-to-digital card then captured the voltage data, with customized software, sampling at 500 Hz, converting the voltage data into displacement data. The BMS system was calibrated against known distances for the range on which the jump squats and bench throws were performed; this calibration was performed before all testing sessions. The reliability of the BMS has been assessed for the measurement of PPO during CMJs (with additional weight) and ballistic bench throws in a study by Alemany et al. (2). In this study, the authors reported intraclass correlation coefficients of 0.93 and 0.96 for peak power obtained during the bench throw and jump squat, respectively.

Statistical Analyses

Following a test for the normality of distribution, data were expressed as the mean $\pm$ SD or median (range). Statistical analyses was carried out using a repeated-measures analysis of variance (ANOVA) to determine whether there was a significant difference between the various CMJs followed by paired *t*-test. Statistical significance was declared at $p \leq 0.05$. A Pearson correlation analysis was used to assess the relationship between strength (3RM) and changes in peak power (12-minute PPO – baseline PPO) following potentiation.

RESULTS

Lower-Body Contrast

Repeated-measures ANOVA revealed a significant time effect ($F = 6.657$, $p = 0.023$). Post hoc analyses found a significant decrease in PPO in the CMJ that was performed ~15 seconds after the preload stimulus ($4,568 \pm 509$ W vs. $4,430 \pm 495$ W, $p = 0.005$; Figure 1a). The PPO was $4,668 \pm 565$ W following 4 minutes of recovery, which was not significantly different from the baseline CMJ ($p = 0.240$; Figure 1a). Following 8 and 12 minutes of recovery, there were significant increases in PPO compared with the baseline CMJ ($4,568 \pm 509$ W vs. $4,862 \pm 485$ W, $p = 0.002$, and $4,568 \pm 509$ W vs. $4,911 \pm 444$ W, $p = 0.001$, respectively; Figure 1a). There was no significant change in PPO following 16 and 20 minutes of recovery compared to the baseline CMJ ($p = 0.265$ and $p = 0.258$, respectively). In addition, the increases in performance at the 8- and 12-minute time points were not significantly different ($p = 0.287$). Percentage change in PPO from baseline can be seen in Table 2.

Upper-Body Contrast

Upper-body results were similar to those reported for lower body. The repeated-measures ANOVA indicated a significant time effect ($F = 81.201$, $p < 0.001$). Post hoc analyses revealed a decrease in PPO during the bench throw performed ~15 seconds after the preload stimulus when compared with the baseline bench throw (856 ± 121 W vs. 816 ± 121 W, $p < 0.001$). Following 4 minutes of recovery, there was no significant difference between the PPO compared to baseline (856 ± 121 W vs. 862 ± 115 W, $p = 0.232$). Similar to lower body, subjects produced significantly greater PPO following 8 minutes (856 ± 121 W vs. 880 ± 130 W, $p = 0.002$) and 12 minutes (856 ± 121 W vs. 903 ± 145 W, $p = 0.000$) compared with the baseline bench throw. In addition, PPO at the 16-minute time point was also significantly greater (856 ± 121 W vs. 863 ± 126 W, $p = 0.014$). Following 20 minutes of recovery, the PPO was similar to that produced during

TABLE 2. Absolute and percentage change in peak power output (PPO) before and after the preload stimulus.*

	Absolute PPO values (W)		Percentage change from baseline (%)	
	Upper-body PPO (W)	Lower-body PPO (W)	Upper body (%)	Lower body (%)
Baseline	856 $\pm$ 121	4,568 $\pm$ 509	—	—
~15 s	816 $\pm$ 121†	4,430 $\pm$ 495†	-4.7 $\pm$ 3.0	-2.9 $\pm$ 3.6
4 min	862 $\pm$ 115	4,668 $\pm$ 565	0.9 $\pm$ 3.2	2.3 $\pm$ 7.2
8 min	880 $\pm$ 130‡	4,862 $\pm$ 485‡	2.8 $\pm$ 3.7	6.8 $\pm$ 7.2
12 min	903 $\pm$ 145‡	4,911 $\pm$ 444‡	5.3 $\pm$ 4.5	8.0 $\pm$ 8.0
16 min	863 $\pm$ 126‡	4,687 $\pm$ 441	0.8 $\pm$ 1.5	3.3 $\pm$ 10.3
20 min	858 $\pm$ 116	4,721 $\pm$ 520	0.3 $\pm$ 2.45	3.8 $\pm$ 10.0

* Values are mean $\pm$ SD.

† Indicates a significant decrease compared to baseline values.

‡ Indicates a significant increase compared to baseline values.

the baseline bench throw (856 ± 121 W vs. 858 ± 116 W, $p = 0.587$). Table 2 shows the percentage change in PPO at each time point from baseline.

Strength Levels

A significant correlation was found between 3RM strength and delta potentiation at the 12-minute time point (PPO at 12 minutes – PPO at baseline) for both the lower ($r = 0.631$, $p = 0.009$) and the upper ($r = 0.590$, $p = 0.004$) body. In addition, there was a significant correlation between 3RM strength and delta potentiation at the 8-minute time point for the lower body only ($r = 0.563$, $p = 0.023$). In terms of relative strength (3RM/body mass), there was a significant positive correlation between lower-body relative strength and delta potentiation at the 12-minute time point ($r = 0.631$, $p = 0.009$) only with no such correlation seen in the upper body ($r = 0.213$, $p = 0.342$).

DISCUSSION

The results of the present study indicate that the optimal recovery to maximize the PAP effect on PPO needs to be between 8 and 12 minutes for the lower body (Figure 1a) and 12 minutes for the upper body (Figure 1b). In addition, the present study highlights the fact that when the explosive activity (CMJ or bench throw) was performed immediately after (~ 15 seconds) the preload stimulus, performance was decreased compared to the same exercise performed with no preload stimulus (Table 2 and Figure 1a,b).

The primary aim of the present study was to determine the optimal recovery period between the preload stimulus and the subsequent explosive activity. Previous studies have used recovery periods ranging from 0 to 18.5 minutes (3, 5, 6, 8, 12, 26) with the majority of the studies using periods of approximately 4 minutes in order to allow for phosphocreatine resynthesis following the preload activity (e.g., 3, 5, 8, 26). In the only other study to directly examine this variable, Jensen and Ebben (18) used various recovery periods (10 seconds and 1, 2, 3, and 4 minutes) and reported no significant difference between the power outputs at any time point after the preload stimulus compared to the power output performed before the preload. However, the authors did report a nonsignificant trend toward an improvement in performance and concluded that beyond 4 minutes of recovery may be needed to get a significant performance improvement. To date, only 2 studies have examined the effectiveness of longer than 5 minutes of recovery between the preload and subsequent exercise (6, 19). Jones and Lees (19) reported no significant change in any of the performance variables measured (i.e., rebound height) following 3, 10, or 20 minutes of recovery following a set of heavy squats compared to the pre-CMJ performed before the heavy squats. However, the sample size of this study was small ($n = 8$), and, as indicated by the author, some trends in the data were present, but the power of the statistical tests was low. Chiu et al. (6) reported that average force, average power, and peak power were significantly greater at 18.5 minutes postactivation compared to 5 minutes postactivation. However, as measurements were taken only at these time points, the optimal recovery period could possibly be anywhere between 5 and 18.5 minutes. The current results help clarify the optimal recovery needed in order to achieve maximal increases in PPO from contrast training in both the upper and the lower body. In terms of lower-body performance, our results in-

dicate that on average between 8 and 12 minutes of recovery is required to achieve maximal increases in PPO that were in the order of ~ 7 – 8% . In addition, optimal recovery during upper-body exercise seems to be around the 12-minute mark with performance increases of $\sim 5\%$. As electromyographic recordings were not obtained in this study, one can only speculate on the potential mechanism for this improvement in performance following PAP. However, 2 primary theories have been proposed to date: (a) the preload stimulus acts to enhance motor unit excitability, possibly affecting a number of processes, such as increased motor unit recruitment, increased motor unit synchronization, decreased presynaptic inhibition, or greater central input to the motor neuron, and (b) enhanced phosphorylation of the myosin light chain (MLC), where the preload causes an increase in sarcoplasmic Ca^{2+} that activates MLC kinase, in turn increasing actin-myosin cross bridging (e.g., 16).

In the present study, we also observed a significant decrease in PPO in both the upper and the lower body when the explosive activity was performed immediately (~ 15 seconds) after the preload stimulus (Figure 1a,b). This finding is in agreement with the finding of Jensen and Ebben (18) who reported decreases in peak ground reaction forces and jump height in squat jumps performed immediately after the preload stimulus (5RM). The preload stimulus has previously been shown to acutely influence the neuromuscular processes responsible for PAP as well as influence the related fatigue mechanism of the muscle. Therefore, the muscle performance following the preload stimulus depends on the balance between the degree to which the muscle is fatigued and excited. In other words, if fatigue outweighs potentiation, a decrease in performance will be seen, and if potentiation outweighs fatigue, an increase in performance will be seen. Therefore, from the results of the present study, it can be concluded that when an explosive activity was performed immediately after the preload stimulus, the fatigue associated with the preload outweighed the potentiation effect (5).

Despite a number of PAP studies showing an ergogenic effect on performance (3, 6, 13, 26), there are a significant number of studies that report no ergogenic effect (e.g., 18). As indicated in the review by Hodgson et al. (16), training history and/or strength levels of the subjects seem to be important factors in the design and outcome of PAP studies. Studies to date have used subjects of varying strength levels (from recreationally trained to power athletes), and in some studies it was only when subjects were differentiated into “strong” and “weak” subjects based on their strength levels (e.g., 3, 15) or training experience (e.g., 6, 14) that a performance effect was observed. For example, Chiu et al. (6) initially reported no change in performance following a preload stimulus when the group were considered as a whole; however, once the group was divided on the basis of strength, performance increases were observed (6). In support of this, the studies by Young et al. (26) and Duthie et al. (8) found significant correlations between performance changes following the preload stimulus and measures of strength (e.g., 1RM; $r = 0.73$ and $r = 0.66$, respectively), which indicates that stronger subjects have greater potential for gains following a preload activity. Results from the present study also show positive correlations between the subjects’ strength (absolute and relative) level and the change in performance following potentiation at the 12-minute time point in both the upper and the lower body, indicating that the

stronger the individual, the greater potential for increase in PPO. This relationship may help explain some of the contradictory findings within the literature. While the exact reason behind this relationship between strength and potentiation remains unclear, it has been demonstrated that resistance-trained athletes have greater activation of the musculature involved during a preload stimulus, which would affect the H-reflex and myosin regulatory light chain phosphorylation the 2 mechanisms involved in the PAP phenomenon (6).

In conclusion, the results from the present study indicate that muscle performance (e.g., power) is enhanced following a preload stimulus in both the upper and the lower body, provided that adequate recovery is given between the 2 activities. In addition, the athlete's initial strength level plays an important role in the ability to utilize this PAP phenomenon.

PRACTICAL APPLICATIONS

The current findings indicate that if coaches are using contrast training methods as a means to increase their players' power, it is advised that they leave between 8 and 12 minutes of recovery between the preload stimulus and the explosive activity in order to obtain the greatest benefit. However, practitioners should be aware that individual responses exist in terms of the optimal recovery for PPO following a preload stimulus, and therefore coaches should individually determine their athletes' optimal recovery for PPO in order to effectively develop their athletes' individual power-generating capabilities.

In addition, the results of the present study have application for incorporation into warm-ups for events that require 1-off explosive attempts, such as the 100-m sprint and the high jump.

REFERENCES

1. AAGAARD, P., E.B. SIMONSEN, J.L. ANDERSEN, P. MAGNUSSEN, AND P. DYHRE-POULSEN. Neural adaptation to resistance training: Changes in evoked V-waves and H-reflex responses. *J. Appl. Physiol.* 92:2309–2328. 2002.
2. ALEMANY, J.A., C.E. PANDOLF, S.J. MONTAIN, J.W. CASTELLANI, A.P. TUCKOW, AND B.C. NINDL. Reliability assessment of ballistic jump squats and bench throws. *J. Strength Cond. Res.* 19:33–38. 2005.
3. BAKER, D. Acute effects of alternating heavy and light resistances on power output during upper-body complex power training. *J. Strength Cond. Res.* 17:493–497. 2003.
4. BAKER, D., AND R.U. NEWTON. Acute effect on power output of alternating an agonist and antagonist muscle exercise during complex training. *J. Strength Cond. Res.* 19:202–205. 2005.
5. BRANDENBURG, J.P. The acute effects of prior dynamic resistance exercise using different loads on subsequent upper-body explosive performance in resistance-trained men. *J. Strength Cond. Res.* 19:427–432. 2005.
6. CHIU, L.Z.E., A.C. FRY, L.W. WEISS, B.K. SCHILLING, L.E. BROWN, AND S.L. SMITH. Postactivation potentiation response in athletic and recreationally trained individuals. *J. Strength Cond. Res.* 17:671–677. 2003.
7. CLARK, R.A., A.L. BRYANT, AND P. REABURN. The acute effects of a single set of contrast preloading on a loaded countermovement jump training session. *J. Strength Cond. Res.* 20:162–166. 2006.
8. DUTHIE, G., W.B. YOUNG, AND D.A. AITKEN. The acute effects of heavy loads on jump squat performance: An evaluation of the complex and contrast methods of power development. *J. Strength Cond. Res.* 16:530–538. 2002.
9. EBBEN, W.P., R.L. JENSEN, AND D.O. BLACKARD. Electromyographic and kinetic analysis of complex training variables. *J. Strength Cond. Res.* 14: 451–456. 2000.
10. FRENCH, D.N., W.J. KRAEMER, AND C.B. COOKE. Changes in dynamic exercise performance following a sequence of preconditioning isometric muscle actions. *J. Strength Cond. Res.* 17:678–685. 2003.
11. GOSSEEN, E.R., AND D.G. SALE. Effect of postactivation potentiation on dynamic knee extension performance. *Eur. J. Appl. Physiol.* 83:524–530. 2000.
12. GOURGOULIS, V., N. AGGELOUSSIS, P. KASIMATIS, G. MAVROMATIS, AND A. GARAS. Effect of a submaximal half-squats warm-up program on vertical jumping ability. *J. Strength Cond. Res.* 17:342–344. 2003.
13. GULLICH, A., AND D. SCHMIDTBLEICHER. Short-term potentiation of power performance induced by maximal voluntary contractions. XVth Congress of the International Society of Biomechanics, 1996. pp 348–349.
14. HAMADA, T., D.G. SALE, AND J.D. MACDOUGALL. Postactivation potentiation in endurance trained male athletes. *Med. Sci. Sports Exerc.* 32:403–411. 2000.
15. HAMADA, T., D.G. SALE, J.D. MACDOUGALL, AND M.A. TARNOPOLSKY. Postactivation potentiation, fiber type, and twitch contraction time in human knee extensor muscles. *J. Appl. Physiol.* 88:2131–2137. 2000.
16. HODGSON, M., D. DOCHERY, AND D. ROBBINS. Post-activation potentiation. *Sports Med.* 35:585–595. 2005.
17. INTERNATIONAL POWERLIFTING FEDERATION. Technical Rules Book of International Powerlifting Federation. Available at: <http://www.powerlifting-ipf.com/IPF-rulebook.pdf>. Accessed October 1, 2005.
18. JESSEN, R.L., AND W.P. EBBEN. Kinetic analysis of complex training rest interval effect on vertical jump performance. *J. Strength Cond. Res.* 17: 345–349. 2003.
19. JONES, P., AND A. LEES. A biomechanical analysis of the acute effects of complex training using lower limb exercises. *J. Strength Cond. Res.* 17: 694–700. 2003.
20. RASSIER, D.E., AND B.R. MACINTOSH. Coexistence of potentiation and fatigue in skeletal muscle. *Braz. J. Med. Biol. Res.* 33:499–508. 2000.
21. REQUENA, B., M. ZABALA, J. RIBAS, J. ERELINE, M. PAASUKE, AND J.J. GONZALEZ-BADILLO. Effect of post-tetanic potentiation of pectoralis and triceps brachii muscles on bench press performance. *J. Strength Cond. Res.* 19:622–627. 2005.
22. ROBBINS, D.W., AND D. DOCHERTY. Effect of loading on enhancement of power performance over three consecutive trials. *J. Strength Cond. Res.* 19:898–902. 2005.
23. SCOTT, S.L., AND D. DOCHERTY. Acute effects of heavy preloading on vertical and horizontal jump performance. *J. Strength Cond. Res.* 18:201–205. 2004.
24. SMILIOS, L., T. PILIANIDIS, K. SOTIROPOULOS, M. ANTONAKIS, AND S.P. TOKMAKIDIS. Short-term effects of selected exercise and load in contrast training on vertical jump performance. *J. Strength Cond. Res.* 19:135–139. 2005.
25. SMITH, J.C., A.C. FRY, L.W. WEISS, Y. LI, AND S.J. KINZEY. The effects of high-intensity exercise on a 10-second sprint cycle test. *J. Strength Cond. Res.* 15:344–348. 2001.
26. YOUNG, W.B., A. JENNER, AND K. GRIFFITHS. Acute enhancement of power performance from heavy load squats. *J. Strength Cond. Res.* 12:82–84. 1998.

Address correspondence to L.P. Kilduff, L.Kilduff@swansea.ac.uk.